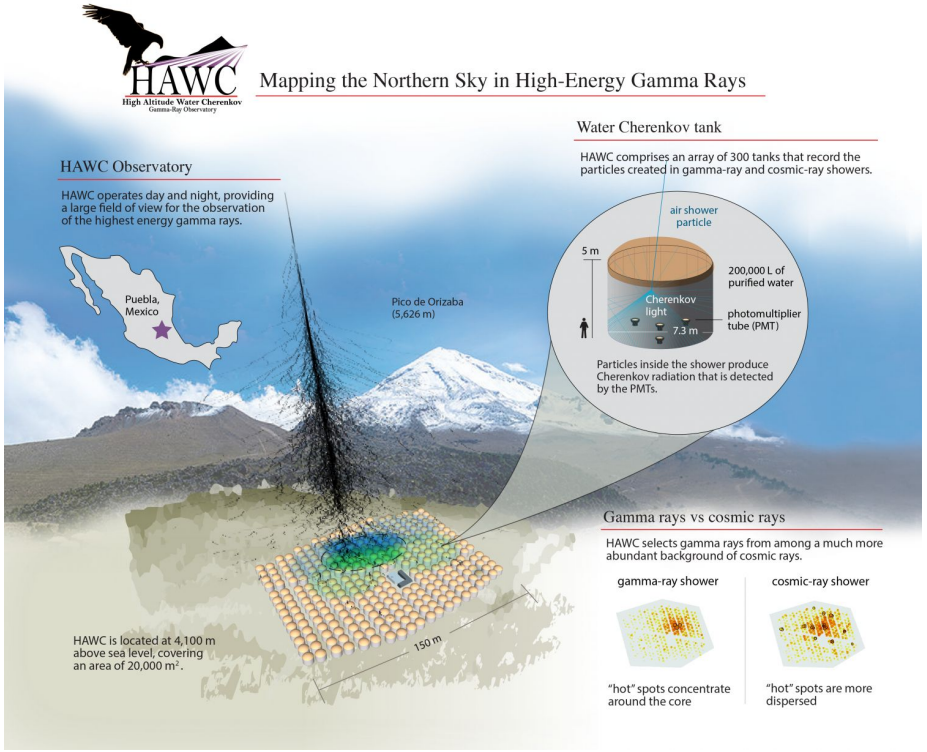
HAWC / IceCube Anisotropy

Riya Kore

University of Wisconsin-Madison



- Background



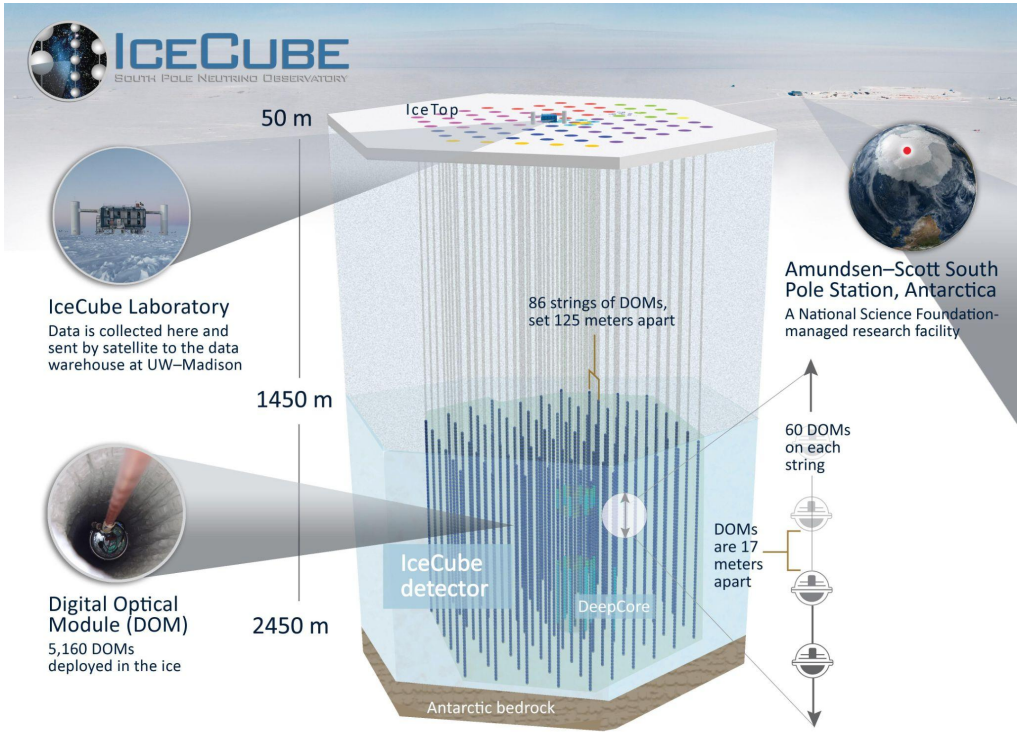
High Altitude Water Cherenkov Observatory

- Located in Mexico
- Sensitive to energy ranges from 100s of GeV to 100s of TeV

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IceCube Neutrino Observatory

HAWC
High Altitude Water Cherenkov
Gamma-Ray Observatory



IceCube Neutrino Observatory

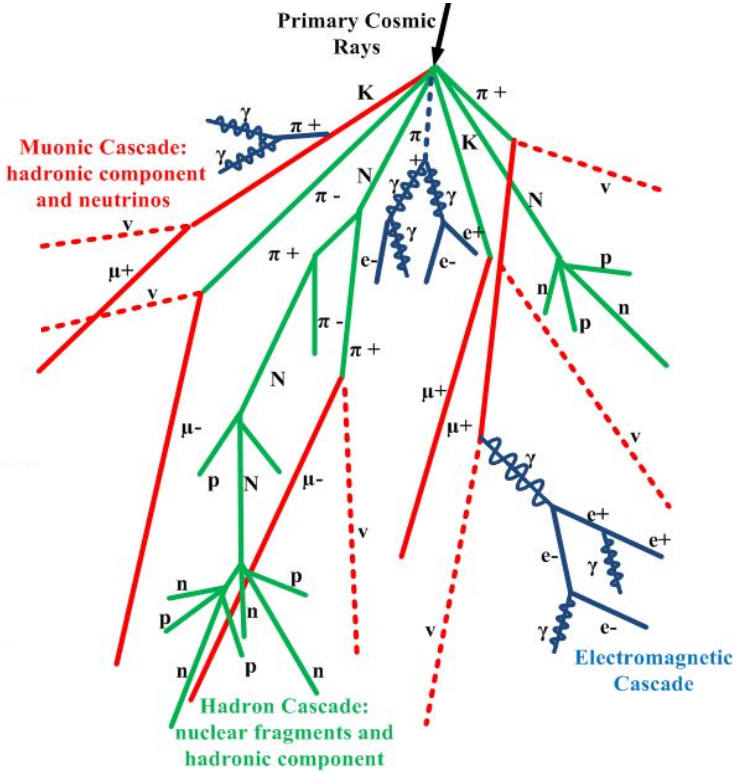
- Located at the South Pole
- Sensitive to energy ranges from 100s of GeV to several PeV

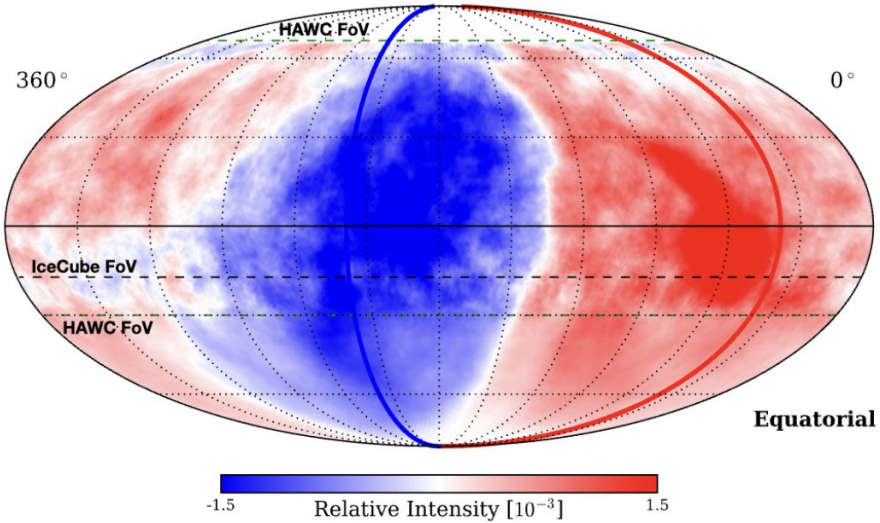
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- Cosmic Rays and Anisotropy





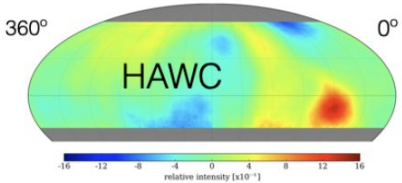
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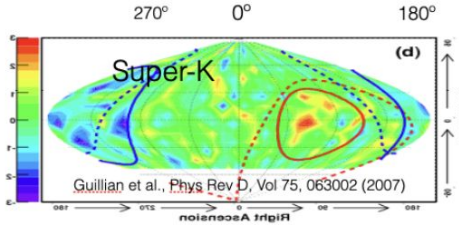
HAWC High Altitude Water Cherenkov Gamma-Ray Observatory

A large-scale ~TeV cosmic ray anisotropy at the level of 10^{-3} has been observed and measured over the last few decades as well as a small-scale structures of angular size from 10° to 30° with an amplitude of 10^{-4}.

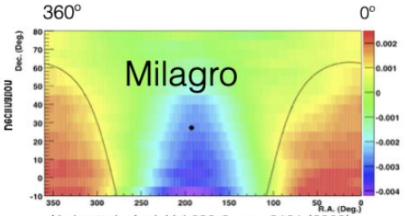
North



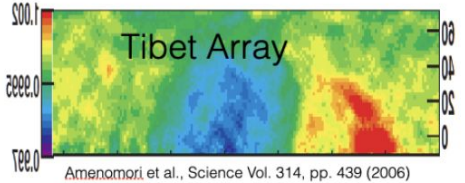
A. U. Abeysekara et al. Astrophys. J. (2014)



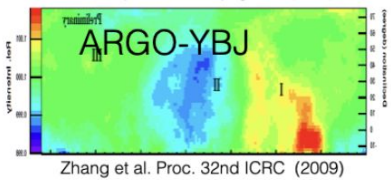
Guillian et al., Phys Rev D, Vol 75, 063002 (2007)



Abdo et al., ApJ, Vol 698-2, pag 2121 (2009)



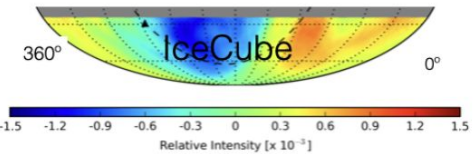
Amenomori et al., Science Vol. 314, pp. 439 (2006)



Zhang et al. Proc. 32nd ICRC (2009)

360° 0° 360° 0°

South



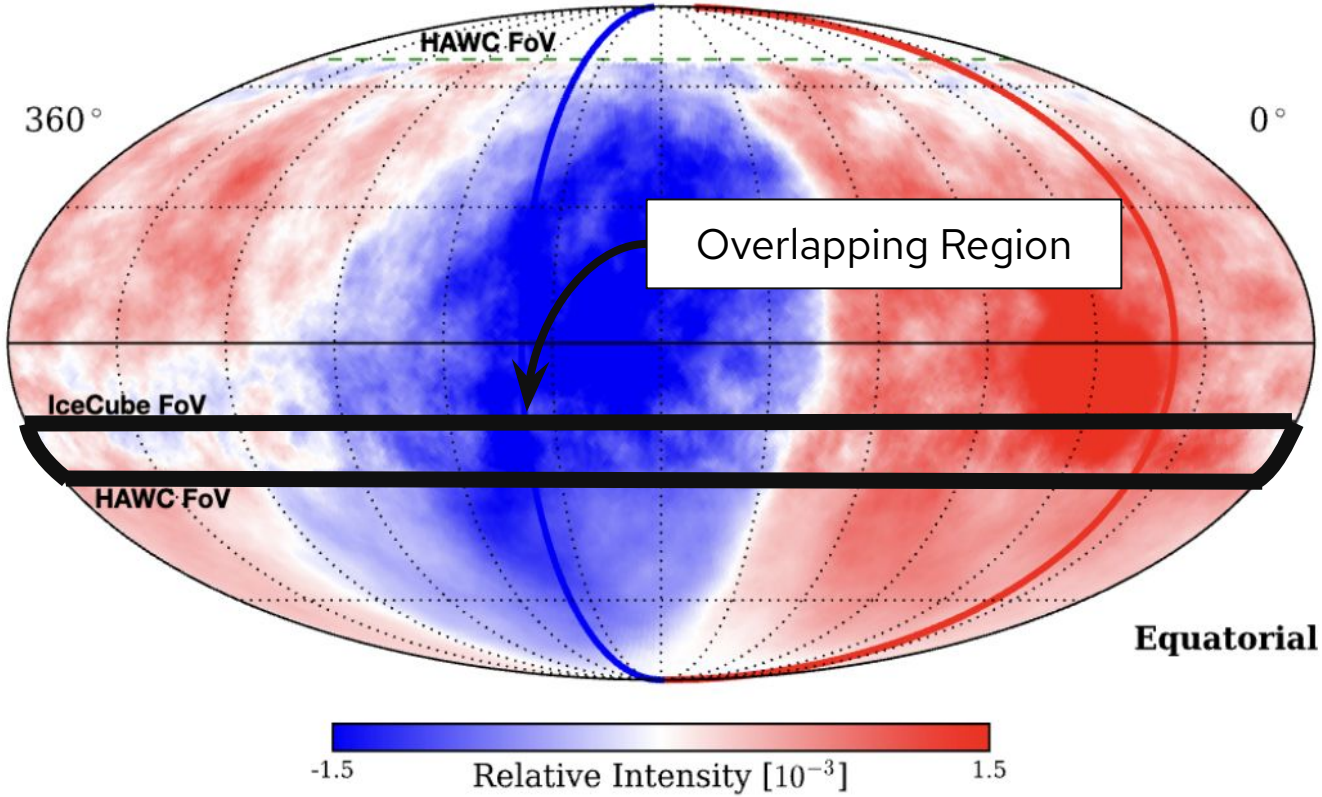
M. G. Aartsen et al. Astrophys.J. 826 (2016)

A number of observatories in the North and only IceCube in the southern hemisphere.

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HAWC High Altitude Water Cherenkov Gamma-Ray Observatory



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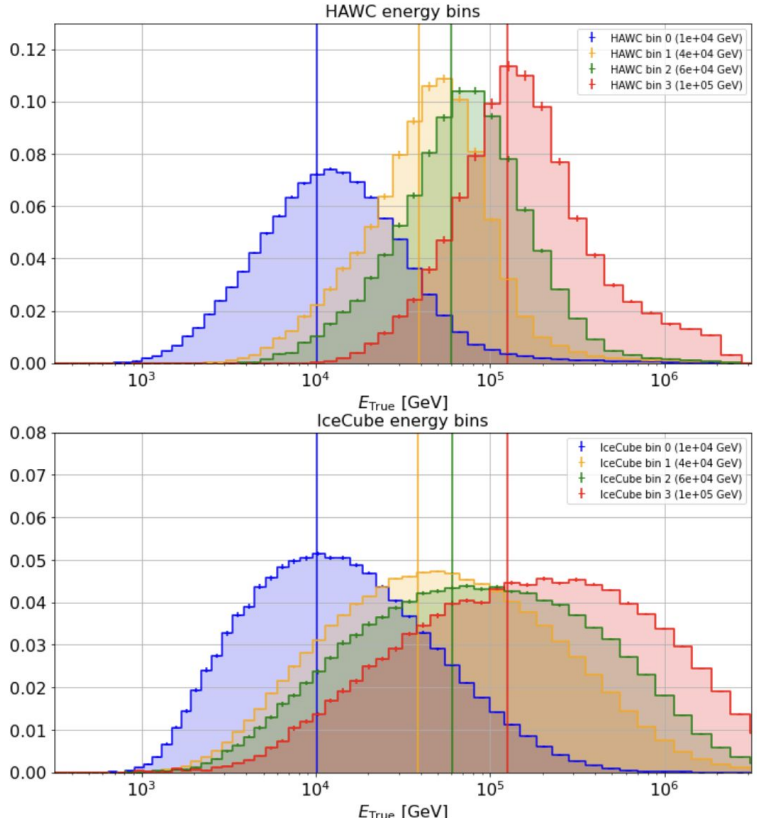
HAWC
High Altitude Water Cherenkov Gamma-Ray Observatory

- What I have worked on

1. Creating **1D distributions of Relative Intensity** in the overlapping region of **HAWC and IceCube**.
2. **Atmospheric Pressure Corrections** in Cosmic Ray Anisotropy measurement with **HAWC**.
3. Binning **HAWC's Monte Carlo Simulation data** to get proper **binning** with respect to the **energy of cosmic ray events**.
4. Redoing the work for **creating 1D sky-maps** with **new data** to get **more accurate results**. (exciting stuff about this later!)

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1. Creating 1D sky-maps of Relative Intensity



These are the **IceCube/HAWC energy bins**

Link to a few papers for more context:
https://arxiv.org/abs/1812.05682
https://iopscience.iop.org/article/10.3847/1538-4357/aad90c

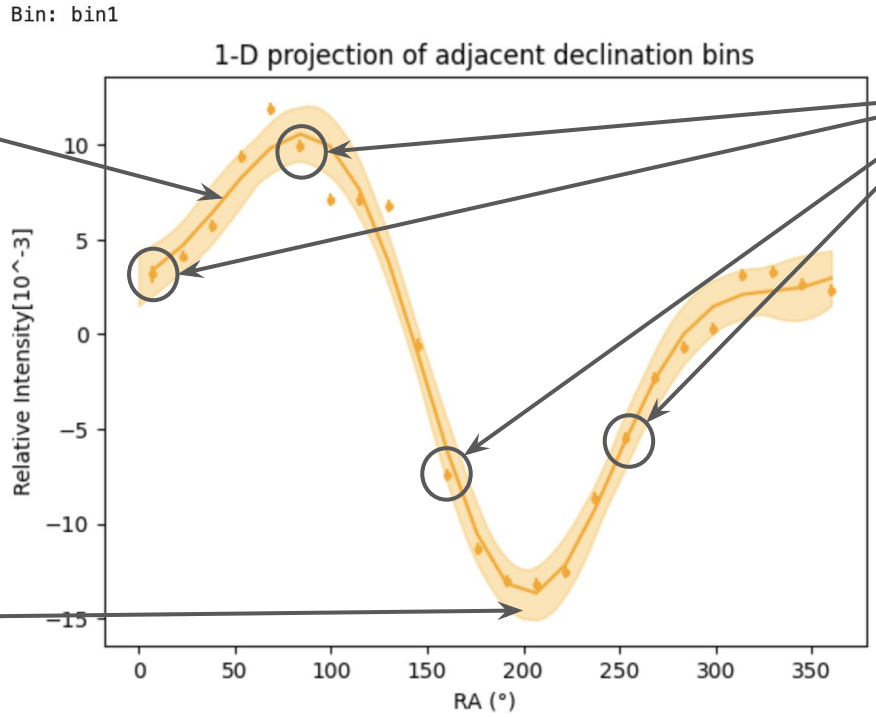
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An example of HAWC's energy bin (Bin 2 out of 4)

The orange line that follows all the points is a function of relative intensity which fits the **dipole + quadrupole + octopole.**



These are basically data points that represent **relative intensity.**

Rel Int = (Data/Background) - 1

This is the **error** for the orange curve fit line.

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My goal was to **plot** the **relative intensities** of **HAWC and IceCube** in the **overlapping region** for the **same energy range** and **visualize the anisotropy**.

Problem 1: HAWC had 4 energy bins while IceCube had 9 energy bins. How can we **match the data** to see the anisotropy?

Solution: **Combine** IceCube bins to match it with HAWC's!

HAWC bin 1 ⟷ IceCube bin 1

HAWC bin 2 ⟷ IceCube bin 3

HAWC bin 3 ⟷ IceCube bin 3 + 4

Bin 1: 10 TeV, Bin 2: 38 TeV, Bin 3: 100 TeV

Problem 2: How do we **check the compatibility** of this data with each other?

Solution: Use a **Chi-squared test** to check for compatibility!

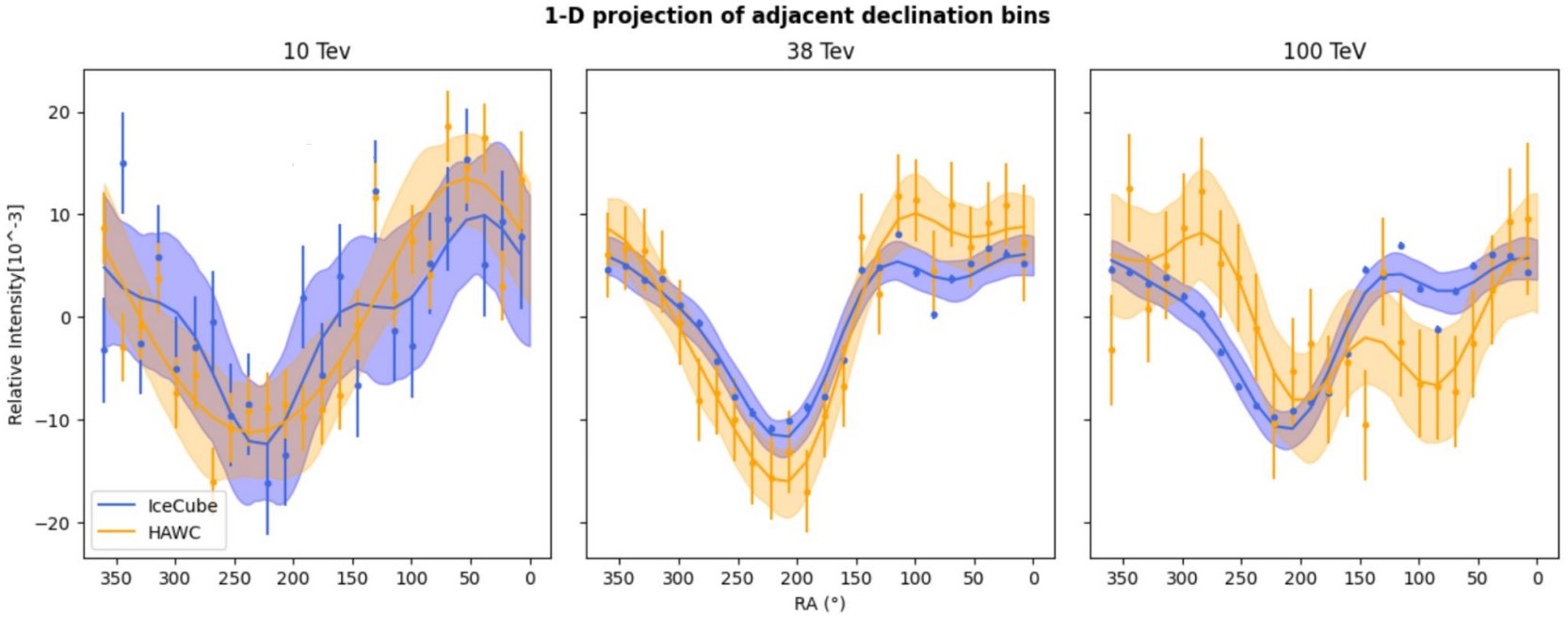
$$\chi^2 = \frac{(\text{rel int } 1 - \text{rel int } 2)^2}{\text{y error } 1 + \text{y error } 2}$$

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Results



$X^2 = 1.824$ $X^2 = 1.061$ $X^2 = 1.893$

We found that **HAWC's bin 2 is most compatible with IceCube's bin 3**.

I presented this at the **Astronomy Club Research Symposium** at UW-Madison in Dec 2023!







2. Atmospheric Pressure Corrections

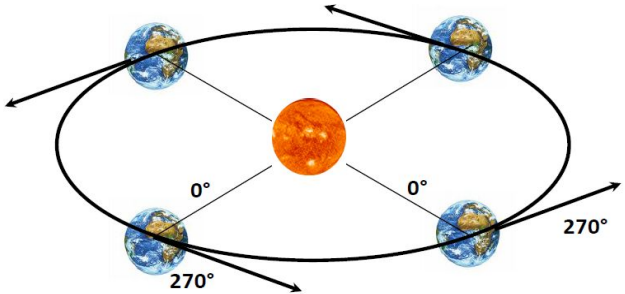
Factors adding to the raw anisotropy:

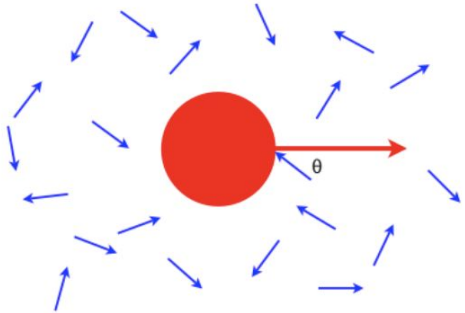
- **Compton Getting Effect -** Difference between the solar and sidereal dipoles caused by the Earth's motion through space.
- **Daily Pressure Variations -** Temperature and Pressure dependence
  - Increase in Pressure reduces detection rates (atmospheric shielding)
  - Decrease in Surface Pressure enhances the detection rates
  - More factors: Lunar Gravitational Tides, Thermally Driven Tides, Weather (seasonal variations)
- **Possible Unknown Factors**

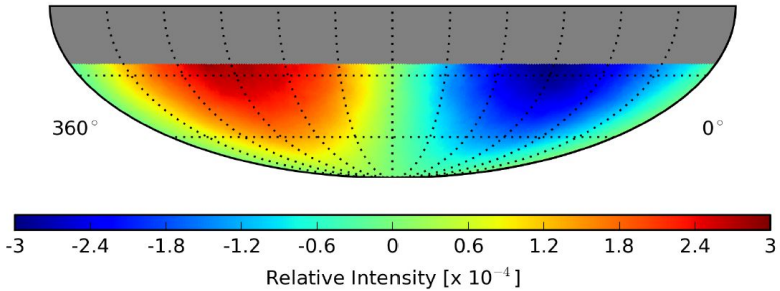
Raw anisotropy + time dependent factors = overall anisotropy detected by HAWC

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Compton Getting Effect







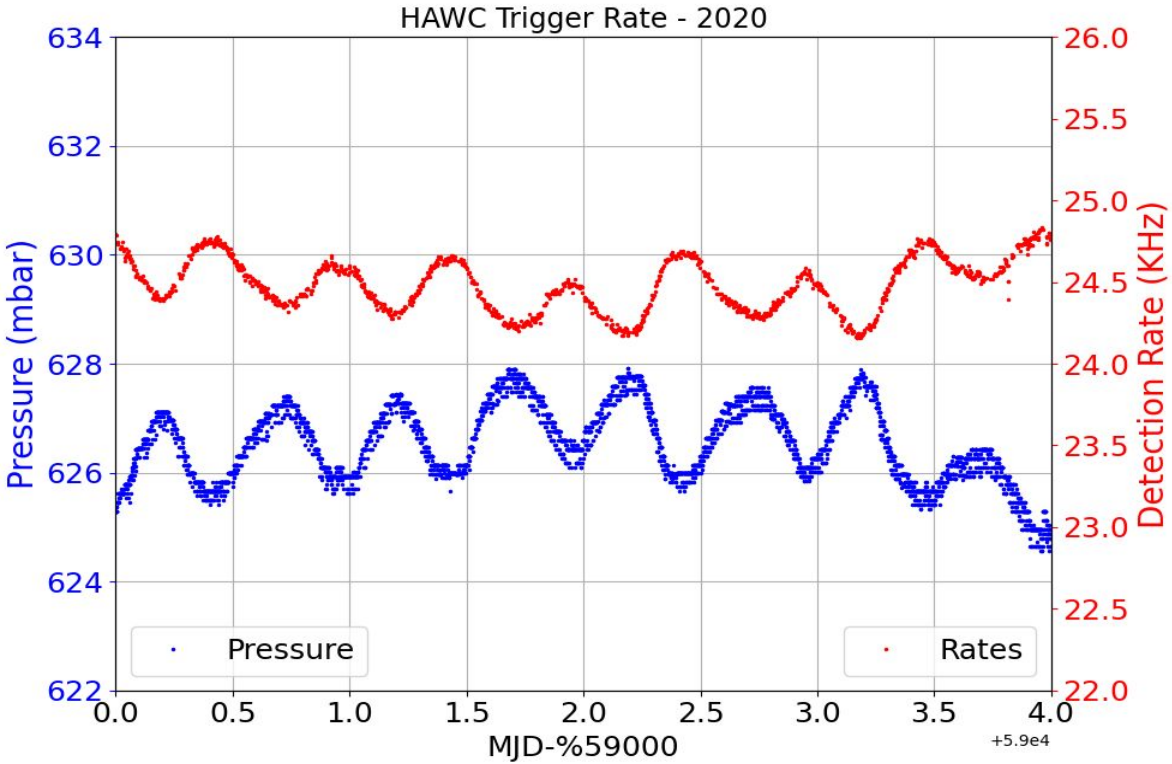
$$\frac{\Delta I}{I} = (\gamma + 2)\frac{v}{c}\cos\theta$$

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Daily Pressure Variations

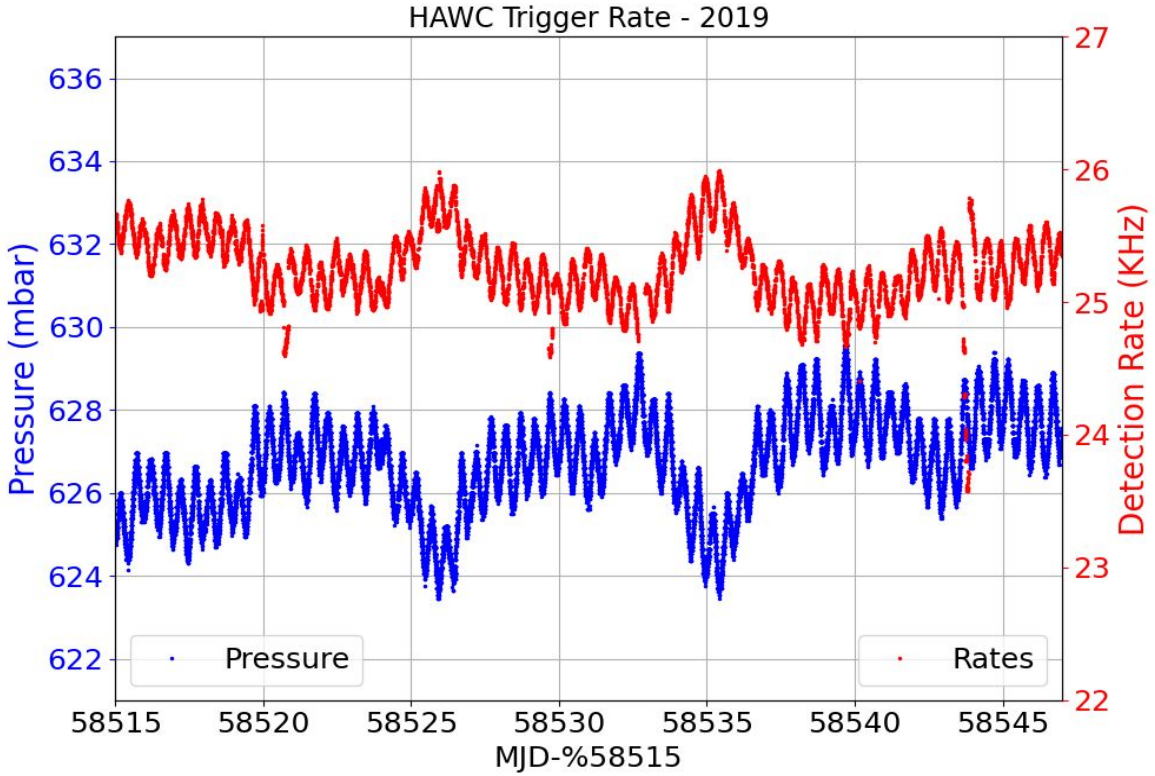


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Monthly Pressure Variations

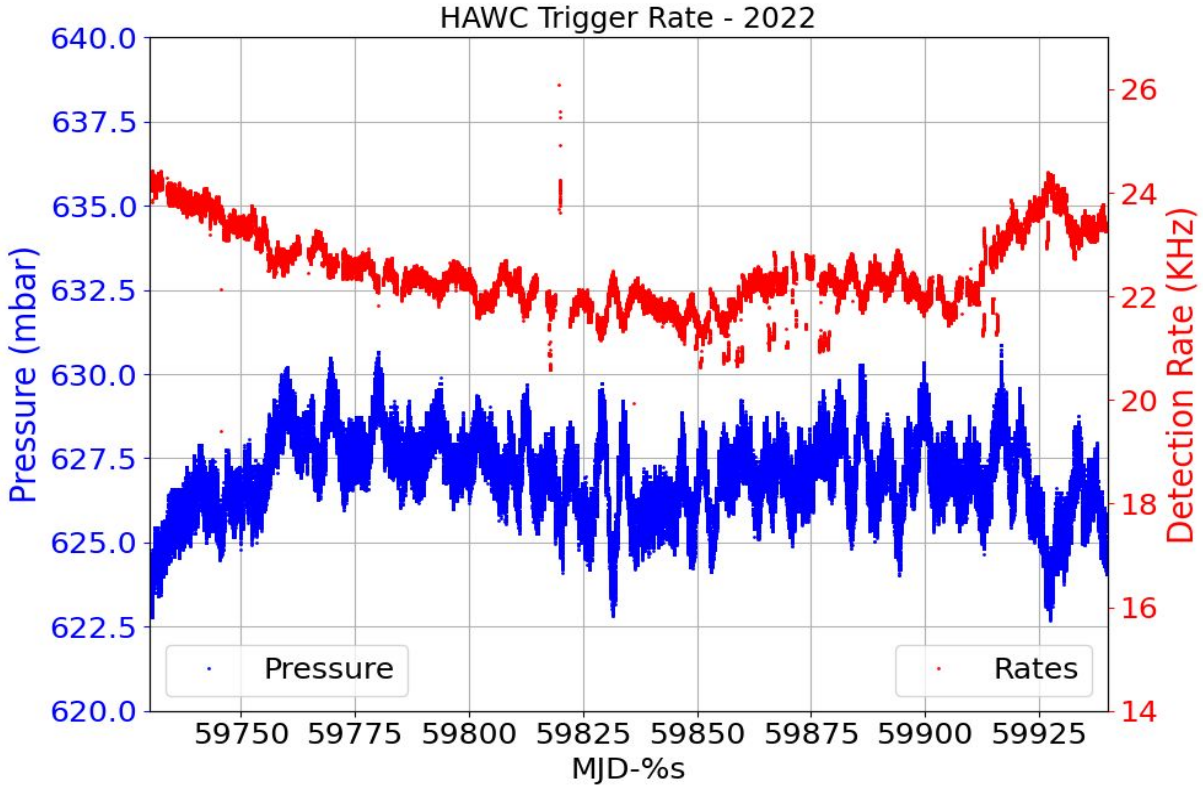


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Possible Seasonal Variations



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The goal was to identify the way to **minimize the time-based pressure dependence** from **detection rates**. It was important to **clean up HAWC's data** to make it **useful for further anisotropy analysis**.

- Used Detection Rates and Pressure data from the years **2015 - 2023**
- Rates data collected every 4 minutes
- Pressure data collected every minute

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1. Calculate the **correlation coefficient** between rates and pressure. Let's call this **β**

2. Calculate the **weight factor**. Let's call this **ω**

3. **Apply** this **weight factor** to the **rates data** to **minimize the pressure dependent** factors.

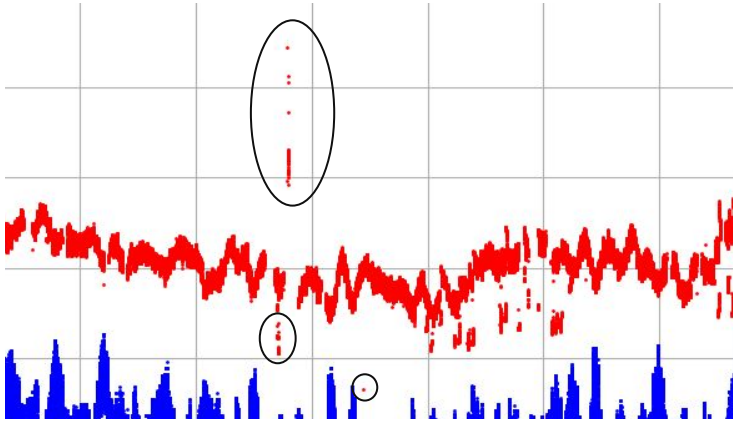
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Cleaning up the data

Problem 1: How to **get rid of** some **values** from the rates data that **go haywire** because of **instrumental errors**?



Solution: **Exclude the rates data** that is **5σ** away **from the mean** for each year.

Problem 2: How to get the **same timestamp** for **rates and pressure data** so as to properly **match** and work with them?

Solution:

1. **Bin** the **pressure and rate timestamps** in **8 minutes** increments.
2. **Apply** this binning to the **rates and pressure** data simultaneously to get the **same timestamp for each bin**.

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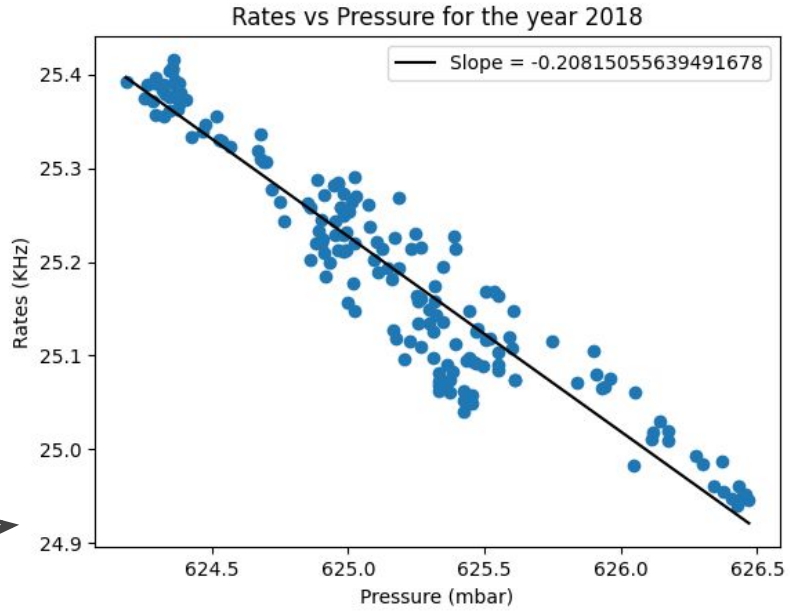
Calculating the correlation coefficient

β = slope of the line of best fit when rates are plotted as a function of pressure

Two methods to get β:

1. Linear correlation
2. Exponential correlation

$$\text{Linear: } \Delta R = \beta_{lin} \Delta P$$

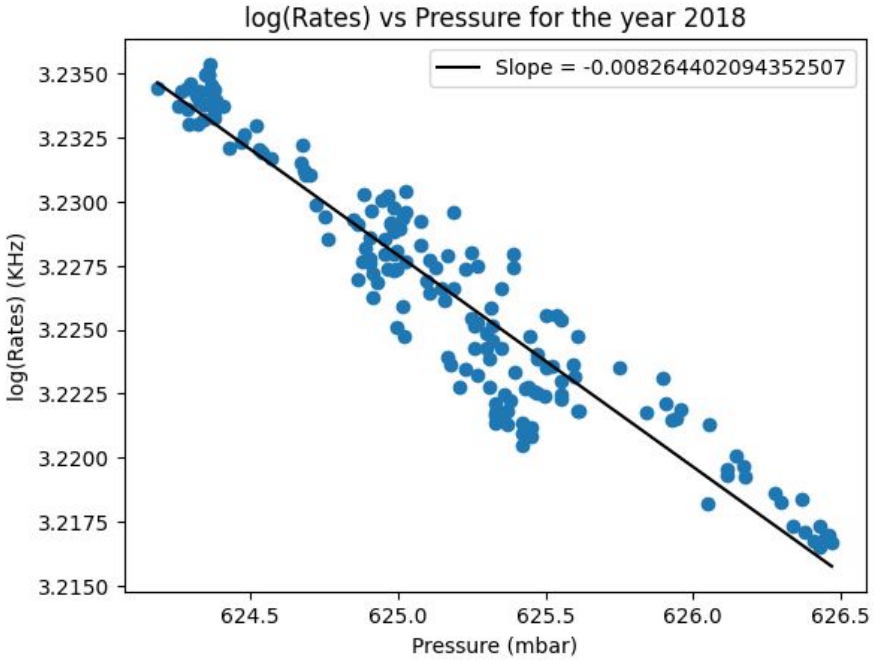


Rates as a function of Pressure for a random day in the year 2018.

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Log(Rates) as a function of Pressure for a random day in the year 2018.

Exponential: $\Delta\{\ln R\} = \beta_{exp}\Delta P$

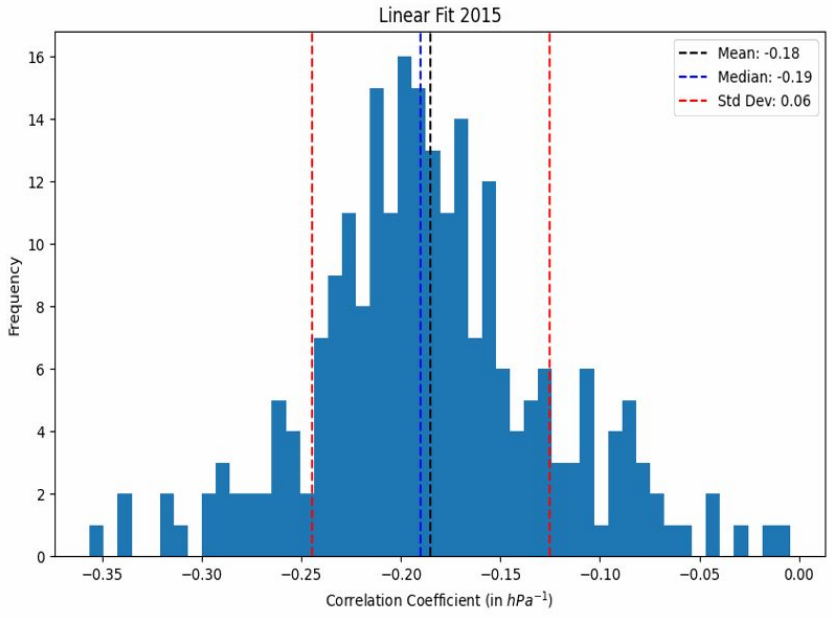
- Calculated this for **each day of each year** using simple Numpy function

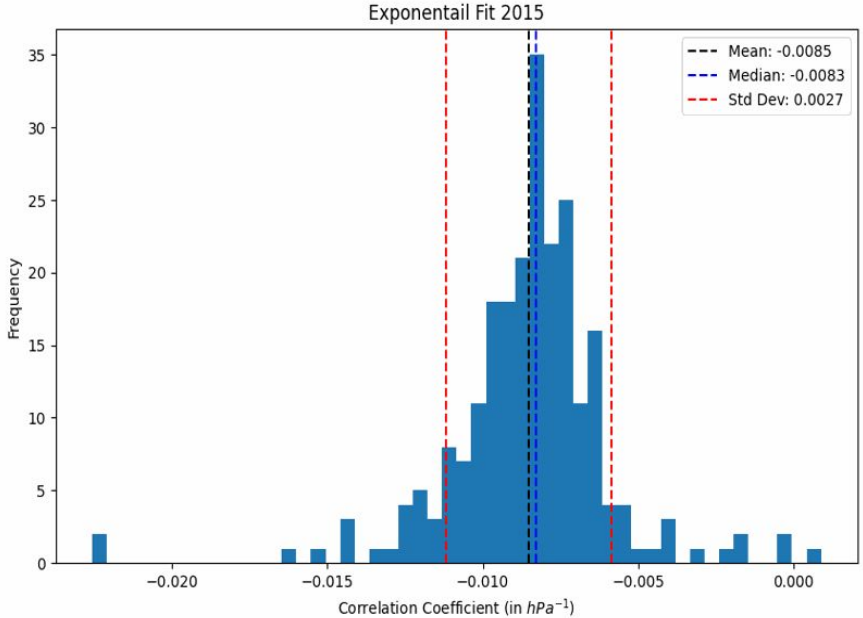
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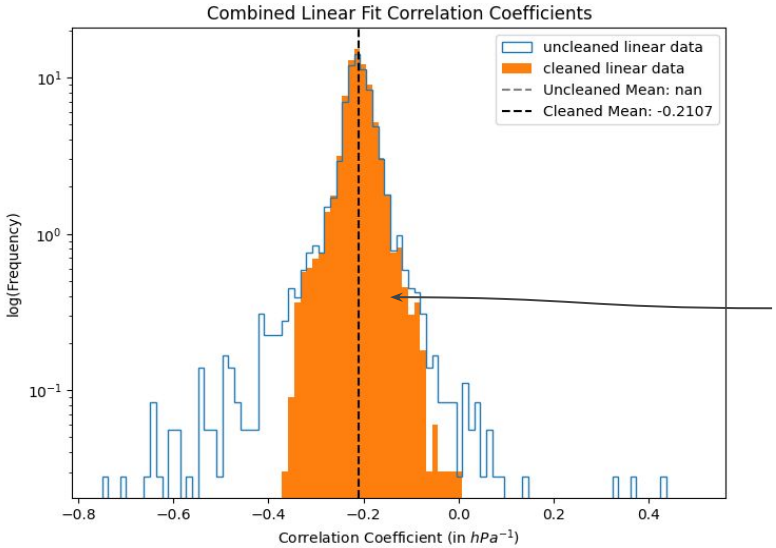
Making β “time independent”



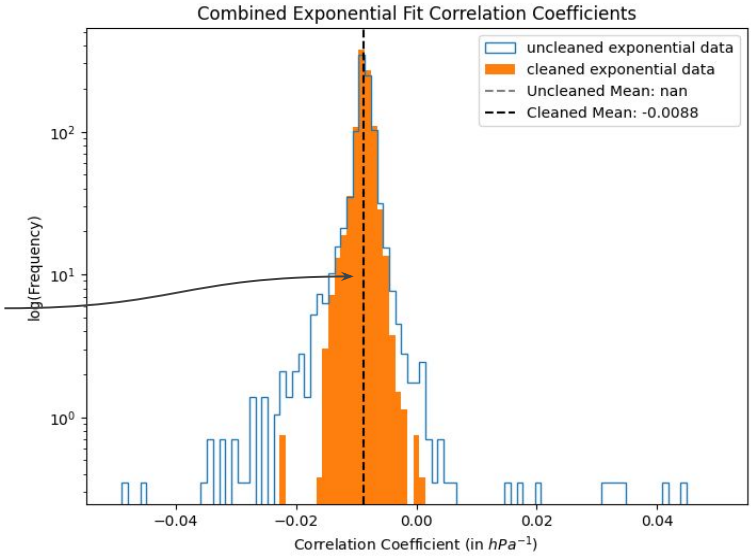


Gaussian Distribution of the linear and exponential correlation coefficients for each year

- Only **considered** the **β values 1σ away** from the **mean** of each year
- **Combined** the values from the cleaned histograms from all years into **one single histogram** (for linear and exponential, respectively)



Cut gives us a more accurate value of the final β

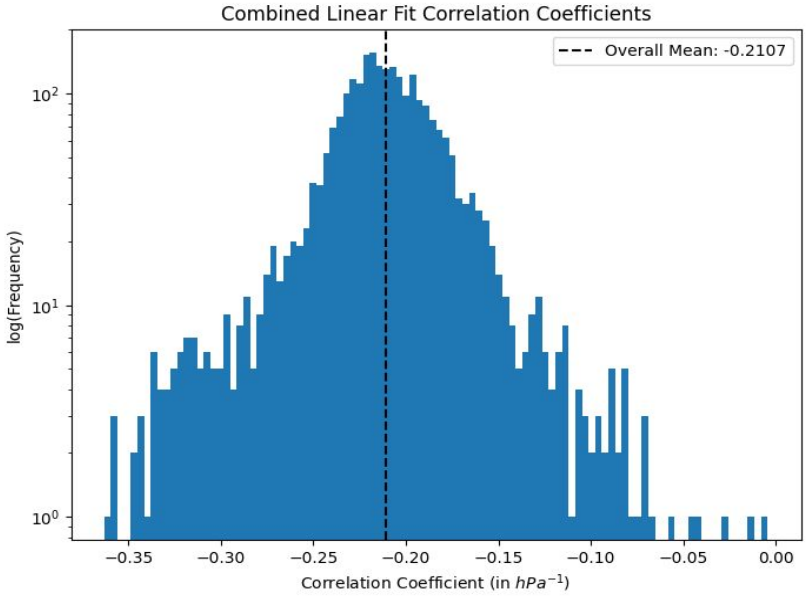


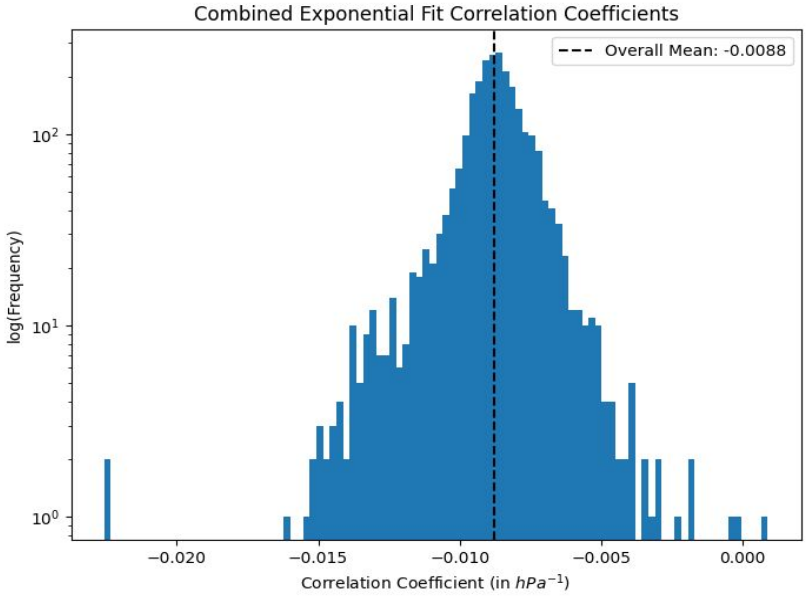
Cut eliminated detector anomalies.

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The mean of these histograms give us the “time-independent” linear and exponential correlation coefficients.

Final β_{lin} = -0.2107 s hPa^{-1}
Final β_{exp} = -0.0088 hPa^{-1}

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1. Calculate the **correlation coefficient** between rates and pressure. Let's call this **β**

2. Calculate the **weight factor**. Let's call this **ω**

3. **Apply** this **weight factor** to the **rates data** to **minimize the pressure dependent** factors.

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Difference between the Linear and Exponential weight factors.

Linear weight factor

$$\omega_{lin} = 1 + \frac{\beta_{lin}}{r_o}(p(t) - p_o)$$

Mean rate for each year in Hz

Mean Pressure for each year hPa

Exponential weight factor

$$\omega_{exp} = e^{\beta_{exp}(p(t) - p_o)}$$

Binned pressure data for that timestamp for each year

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1. Calculate the **correlation coefficient** between rates and pressure. Let's call this **β**

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3. **Apply** this **weight factor** to the **rates data** to **minimize the pressure dependent** factors.

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Correcting the Rates

Multiplying the **weight factor** with the binned rates data to get the **corrected rates**

Linear “corrected rates”

$$rcorr_{lin}(t) = r(t) \times \omega_{lin}$$

Exponential “corrected rates”

$$rcorr_{exp}(t) = r(t) \times \omega_{exp}$$

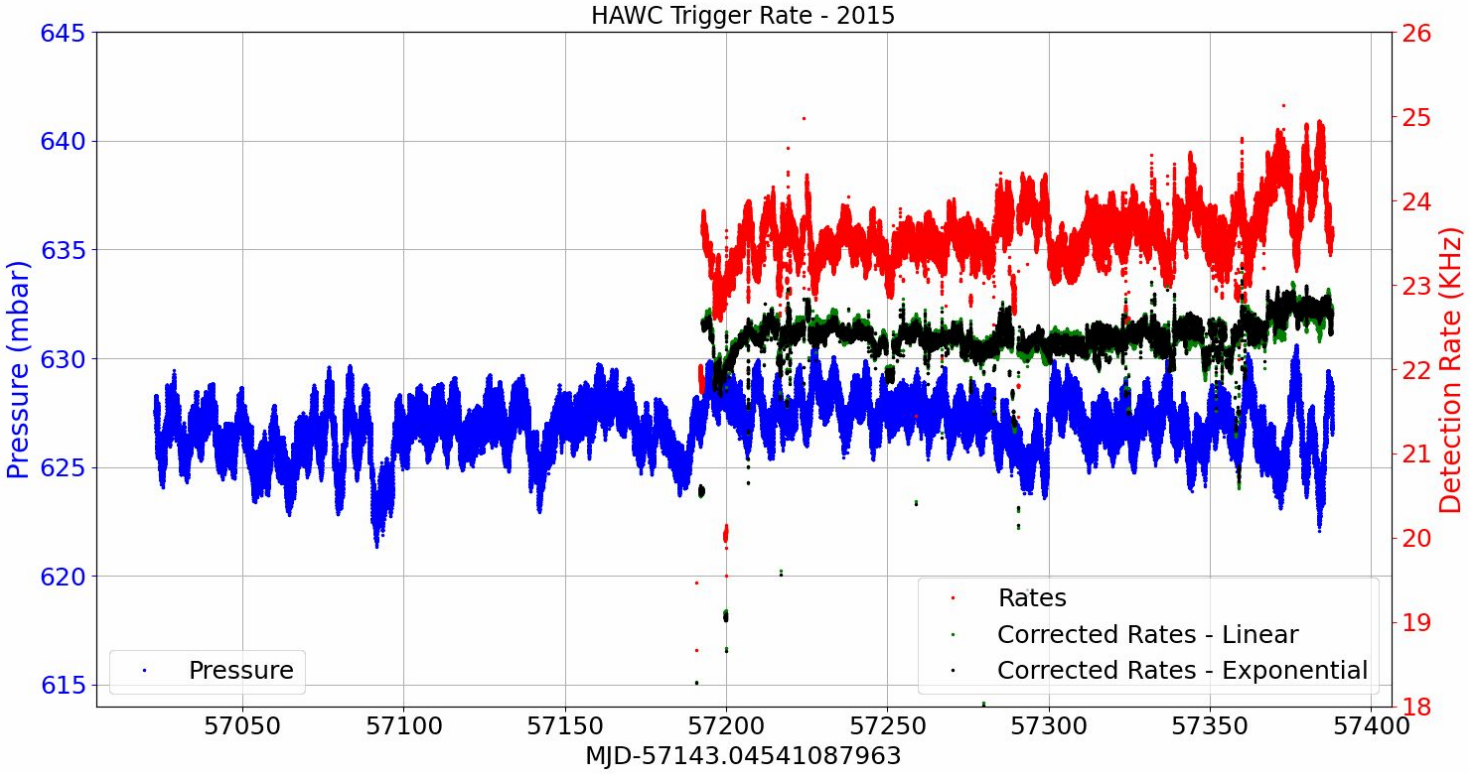
The aim was to **minimize the pressure dependent factors** affecting the rates data.

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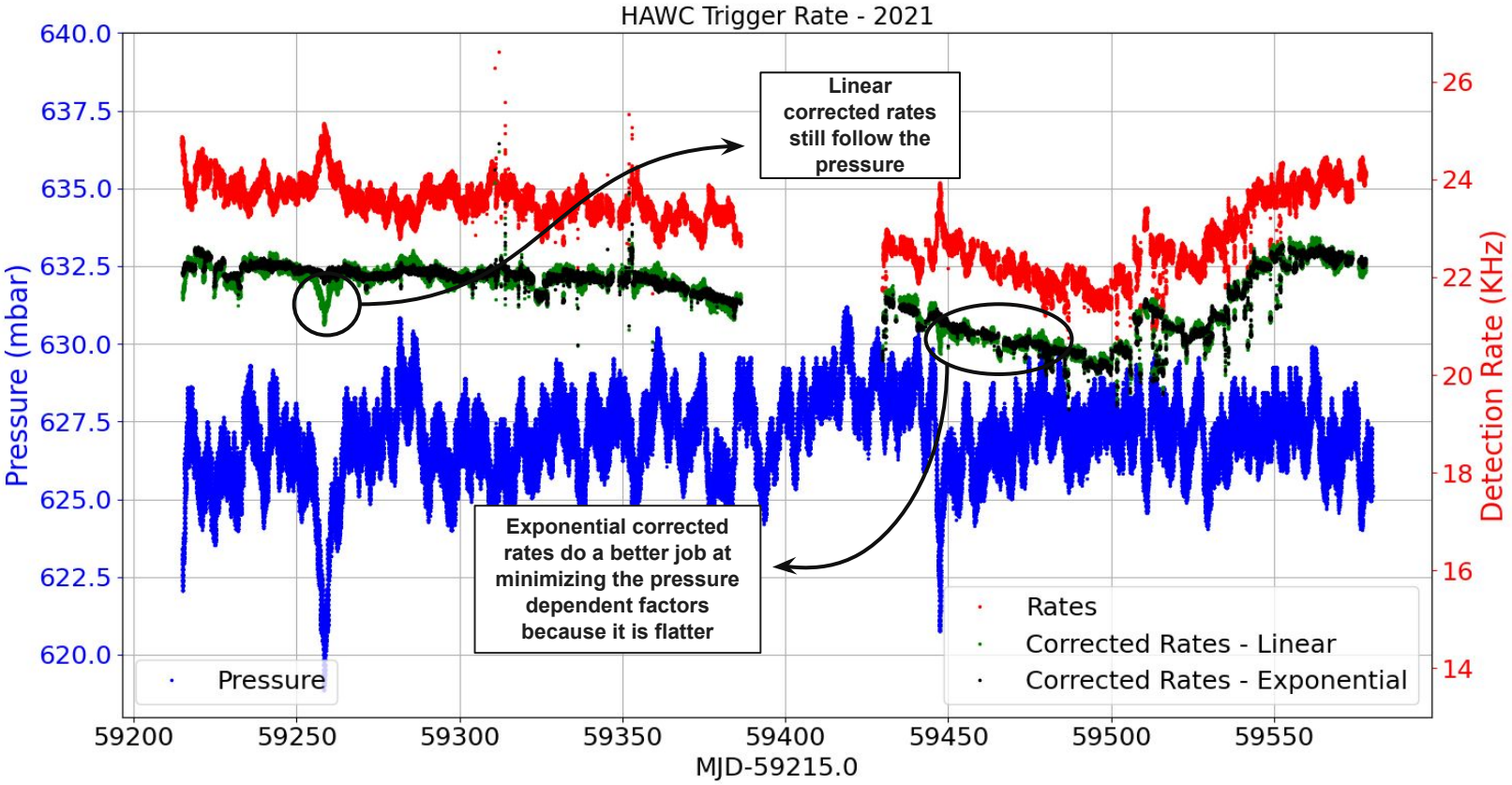
Results



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Conclusion: This is why we went forward with the exponential fit to extract the raw anisotropy.

$$w_i = e^{-\beta(p(t_i)-p_0)}$$

Pressure Correction

$$\delta_\odot = \frac{v(t)}{c}\left[\gamma(E)+2\right]\cos\xi$$

Compton Getting Effect Correction

$$w_{ij} = e^{-\beta(p(t_j)-p_0)}\left(1-\frac{v(t_j)}{c}\left[\gamma(E_j)+2\right]\cos\xi_j\right)$$

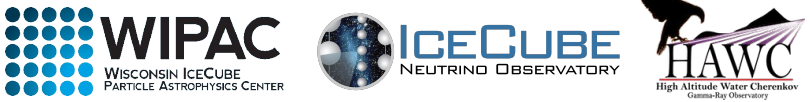
Combined weight factor with both corrections

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I presented this at the **Mid-America Regional Astrophysics Conference** at University of Kansas and the **Astronomy Club Research Symposium** in Dec 2024!



3. Binning Monte Carlo Simulation data for HAWC

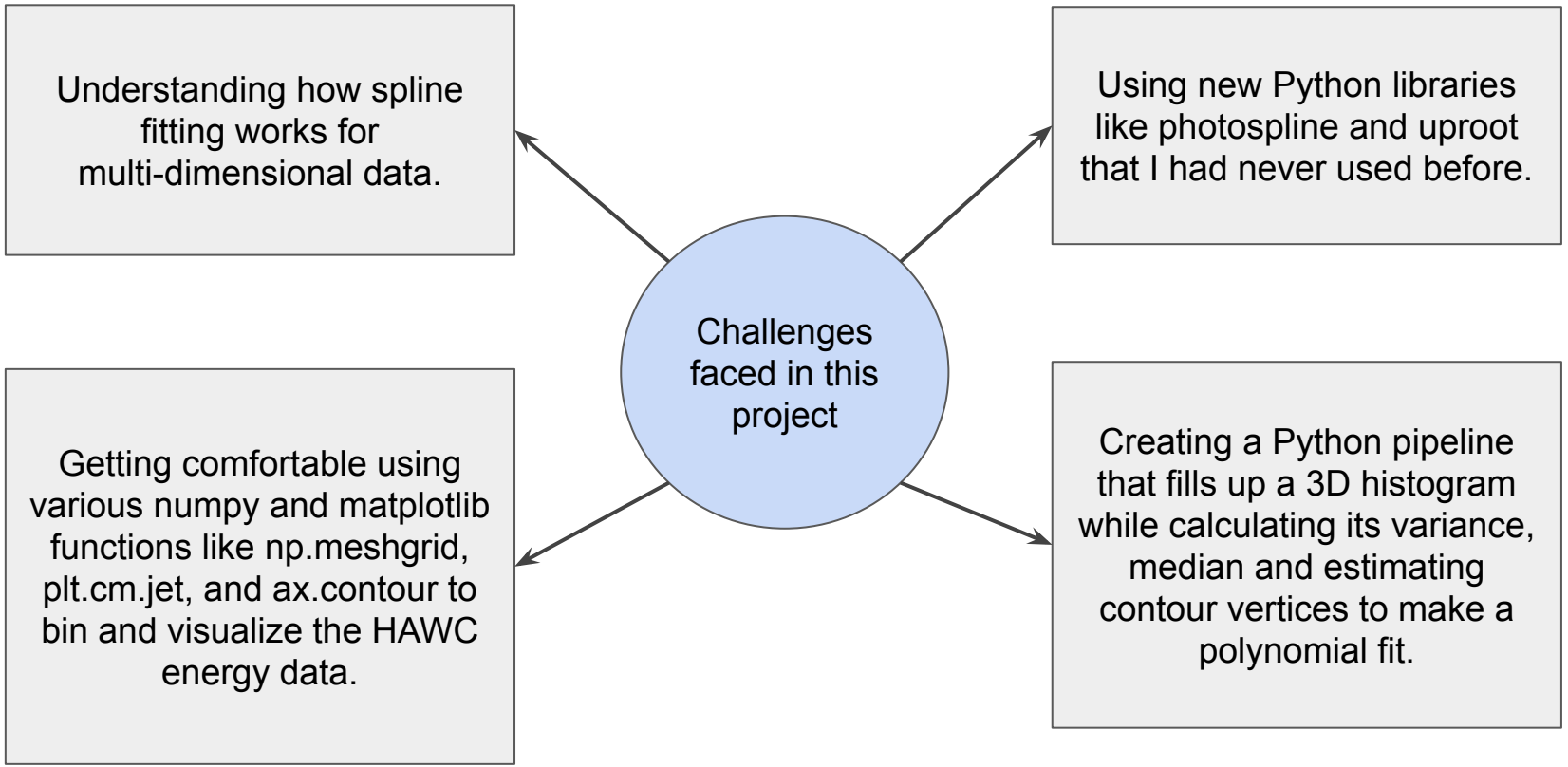
- Initially, **only 4 HAWC energy bins** for the all-sky anisotropy analysis.
- Wanted **more evenly distributed energy bins** that will help visualize the anisotropy better and help with combining HAWC and IceCube data.

Plot a 3D histogram of **log(nHit) vs. cos(zenith)** on the x-y axis and **log(energy)** on the z axis → **Generate splines** with median monte carlo energy → **Apply** the **energy cuts** to get HAWC energy bins

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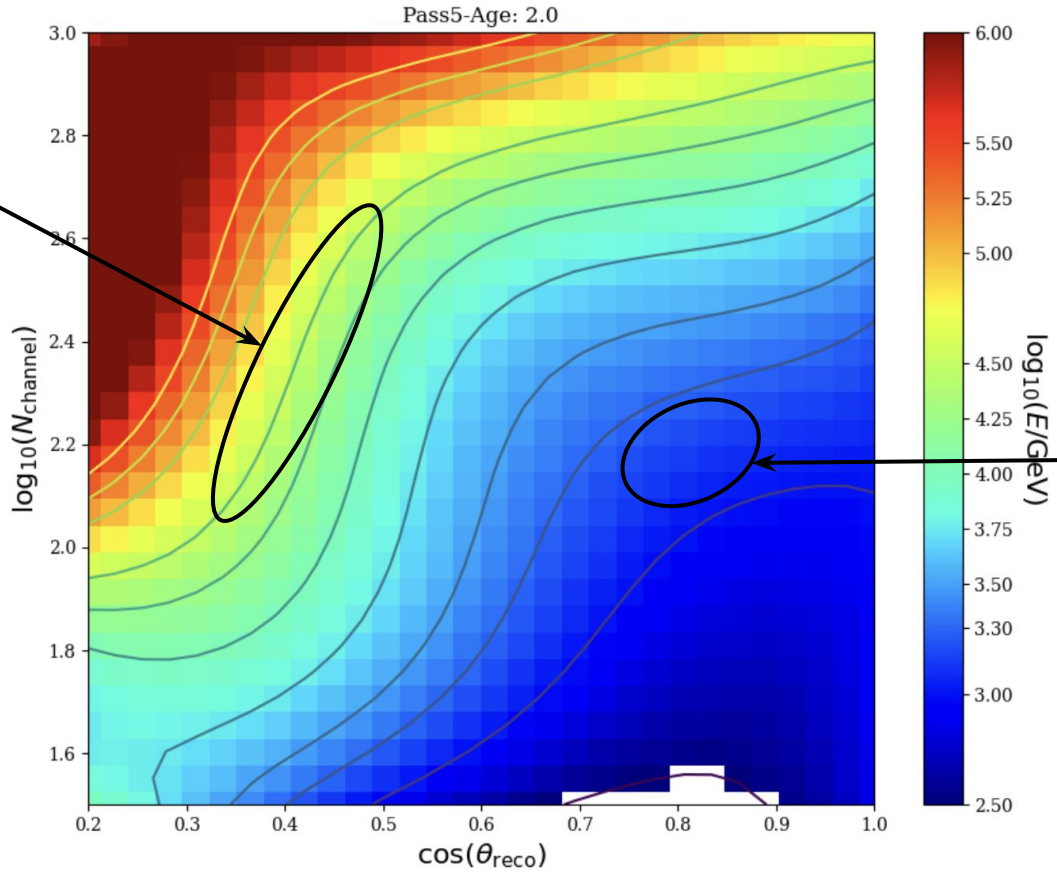


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Polynomial fit lines for the **contour vertices** based on **energy histograms**. These lines represent **HAWC energy cuts**.



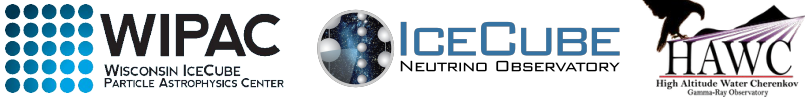
Space between the cuts represent the **events at that energy**.

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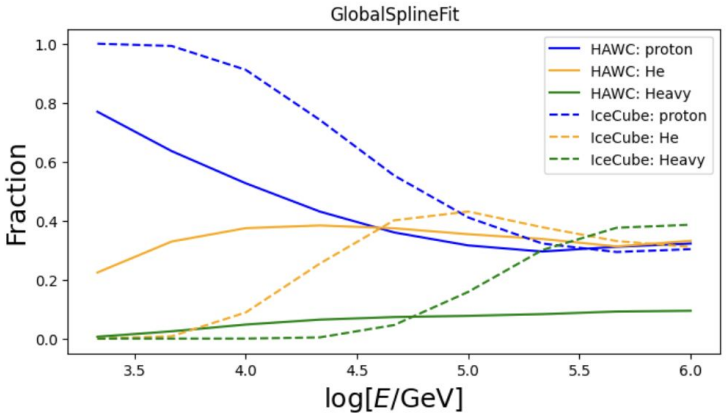
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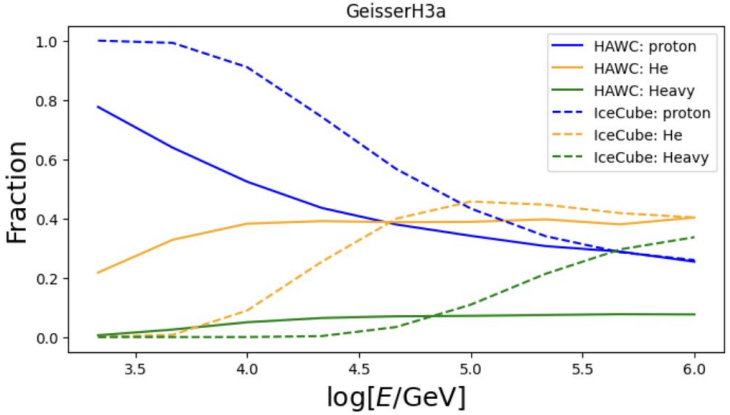
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I can do a quick code walk through if needed…



4. Combining HAWC and IceCube data to get more accurate all sky-maps





Two different **composition assumptions** used here.

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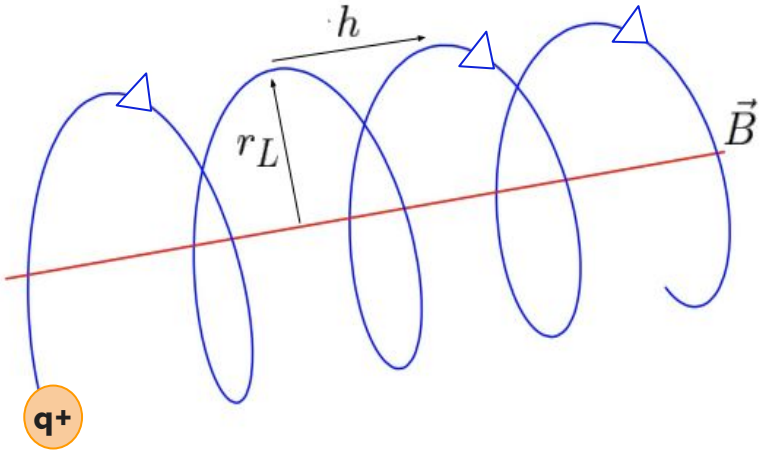
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Magnetic Rigidity

- Path because of **Lorentz Force** $\vec{F} = q(\vec{v} \times \vec{B})$
- **Larmor Radius** (r_L) is the rigidity of the charged particle
- $\text{Rigidity} = mv/q\vec{B}$ for non-relativistic particles
- For Cosmic Rays:

  $\text{Rigidity} = E/Z$



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Hypothesis

- **Cosmic Ray Anisotropy** is a **function of Rigidity**.
- **HAWC and IceCube** have **different sensitivities** to Cosmic Ray **mass** and **charge**, and that is why we need to **scale with Rigidity**.

My goal for this project was to **implement this approach**, **test** it on the **simulation data** and then **apply** it to the **real data** and **generate the all sky maps!**

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1. **Binned** the **per event Rigidity** for HAWC in terms of Energy.

2. Performed the **KS Test** between HAWC and IceCube bins to **compare rigidities.**

3. Got the **most compatible energy ranges** based on the **KS Statistic**.

4. **Made the all sky-maps** for those energy ranges **using** the **real data.**

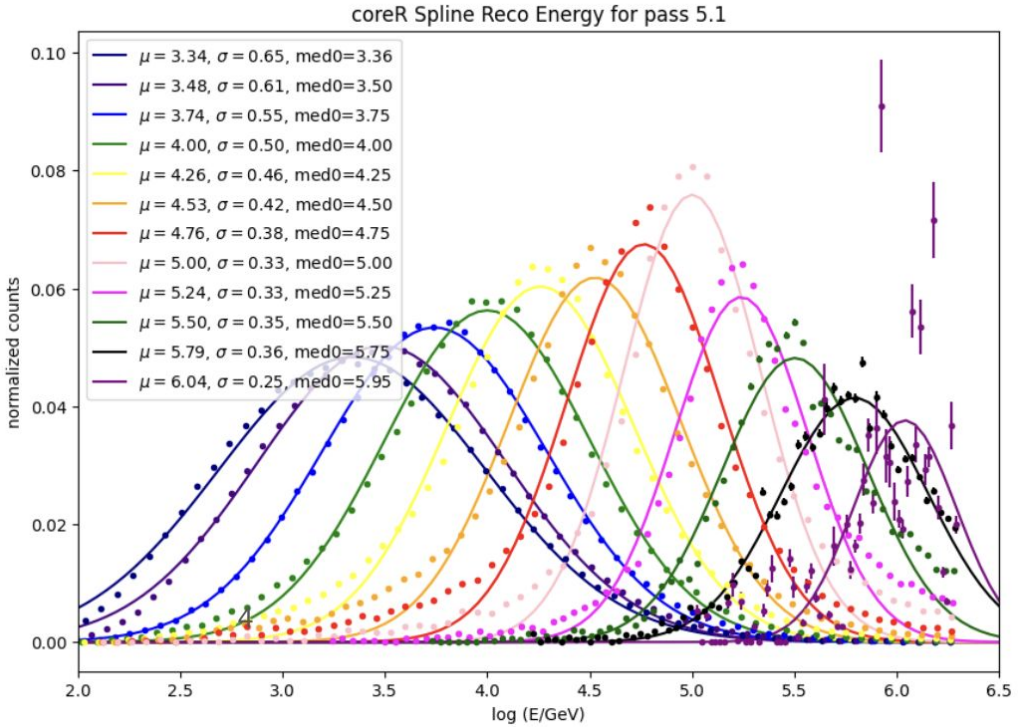
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- The **12 HAWC energy bins** obtained from the previous analysis.

- Used the simple **Rigidity = E/Z** equation to convert energy to rigidity.



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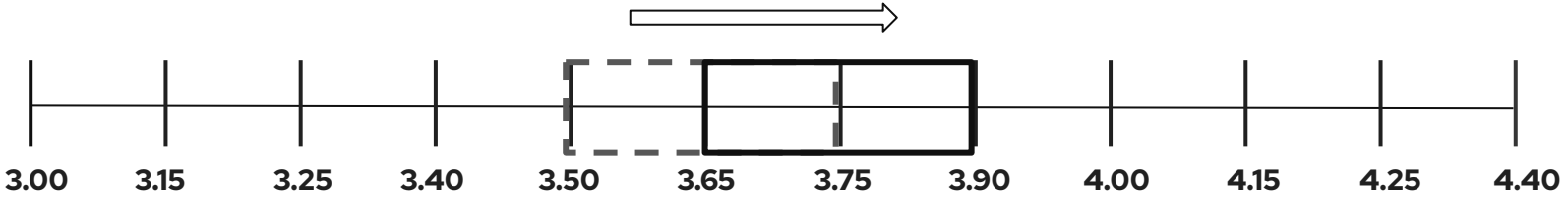
4. **Made the all sky-maps** for those energy ranges **using** the **real data.**

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1. Implemented a **sliding window approach** on IceCube energies.

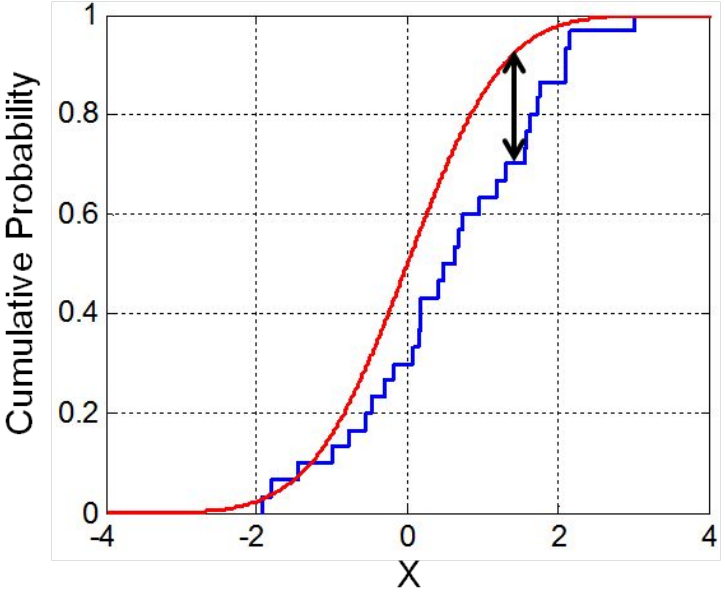


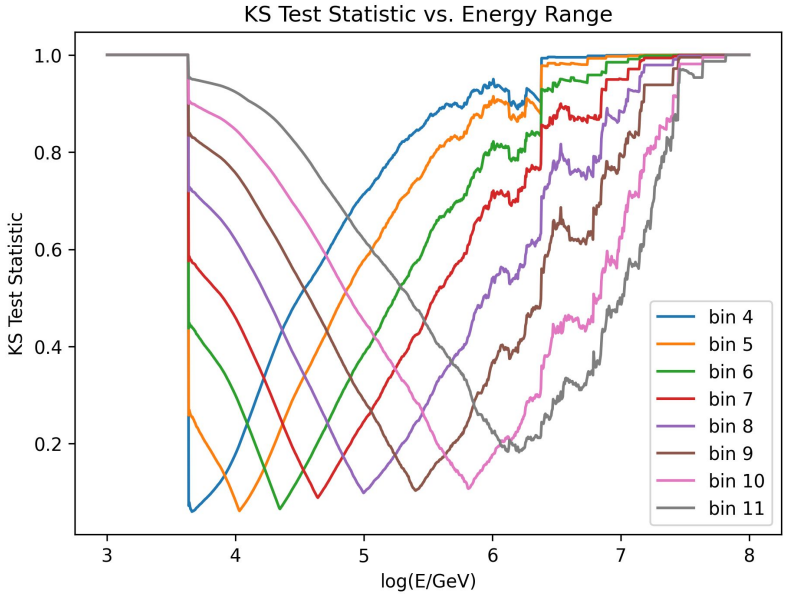
an example of the sliding window approach

2. Calculated the **weighted CDFs** for HAWC and IceCube Rigidities and found the **maximum distance between them**.

$$F(x) = \frac{1}{W} \sum_{i=1}^{n} w_i \{d_i \leq x\}$$

3. Found the **minimum of those distances**. (KS Test)





This gave me the **IceCube energy windows most compatible with HAWC** in terms of Rigidity.

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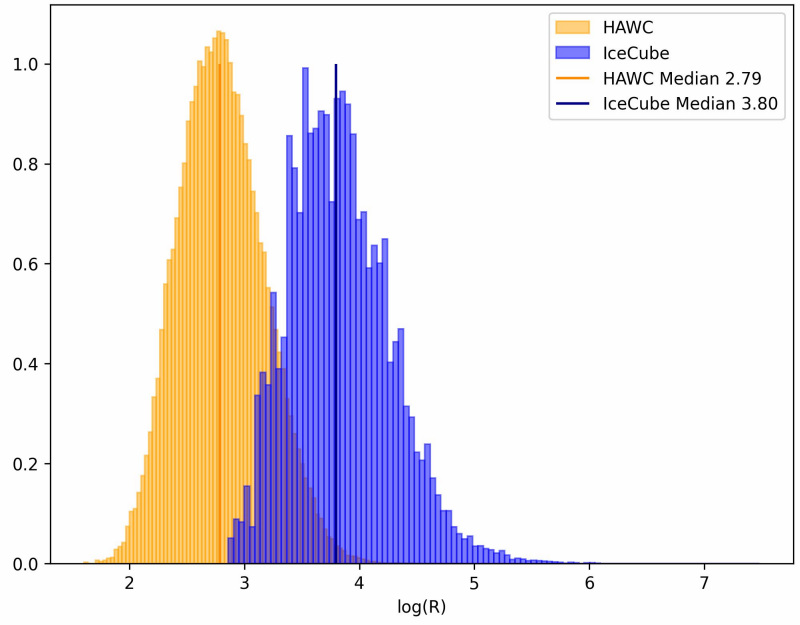
4. **Made the all sky-maps** for those energy ranges **using** the **real data.**

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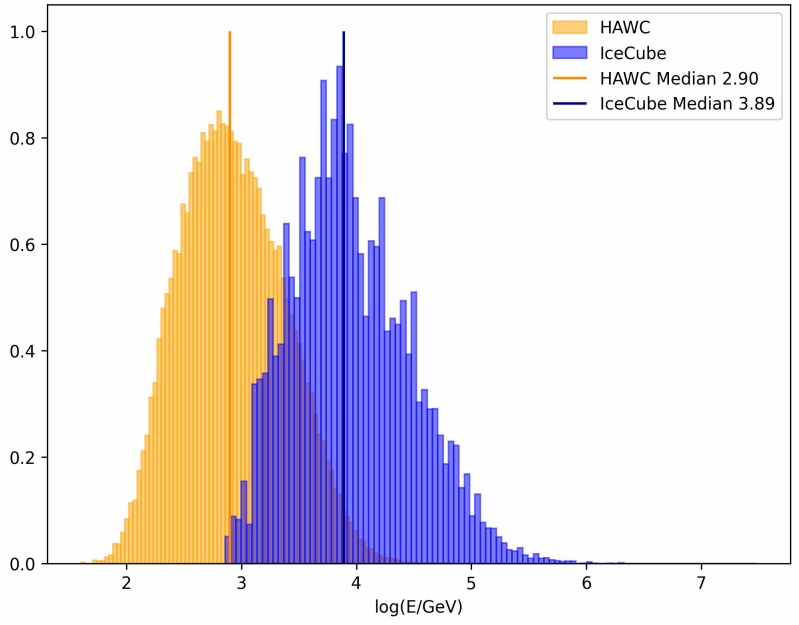
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Results from the simulation analysis!



Rigidity



Energy

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1. Calculate the **Relative Intensity**.

$$\text{Relative Intensity} = \frac{\text{Data}}{\text{Background}} - 1$$

2. **Chi-squared Test**: For a given HAWC map, iterate through the IceCube maps and find the energy bin with the lowest χ^2

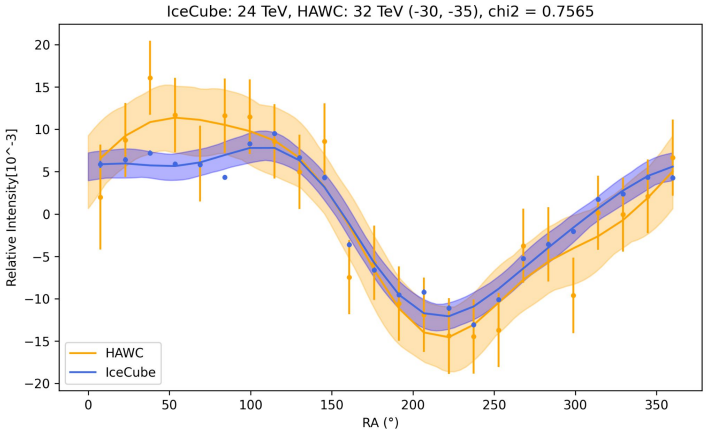
$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

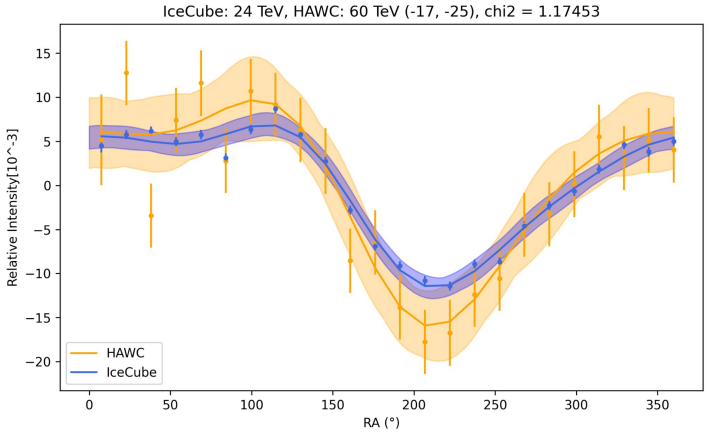
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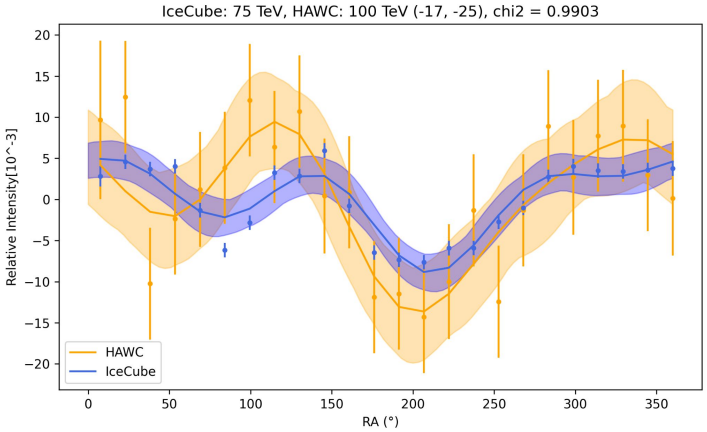
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Preliminary Results





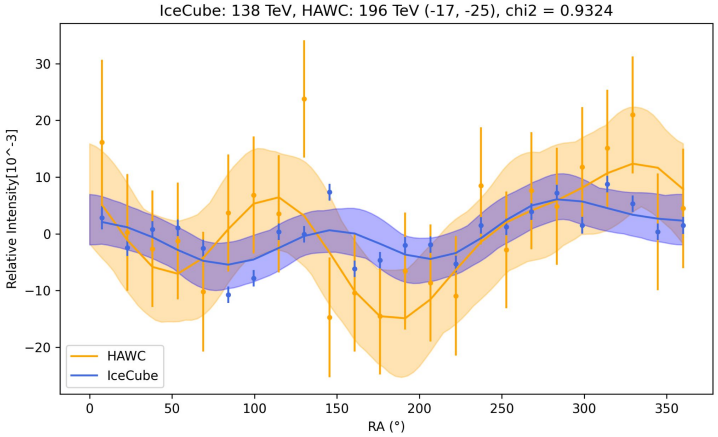


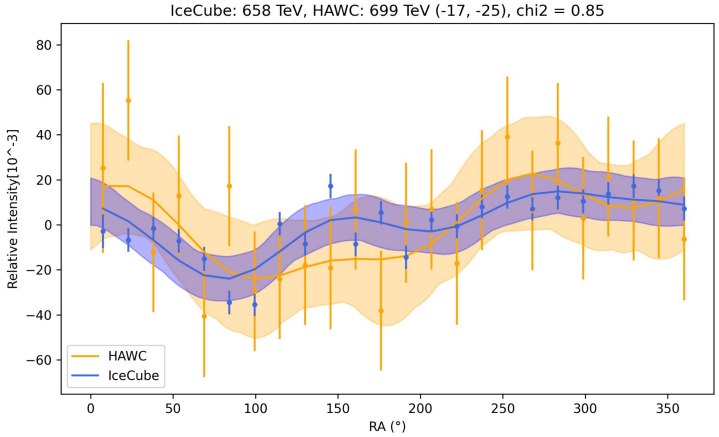
1D projection of the All Sky Maps of the real data.

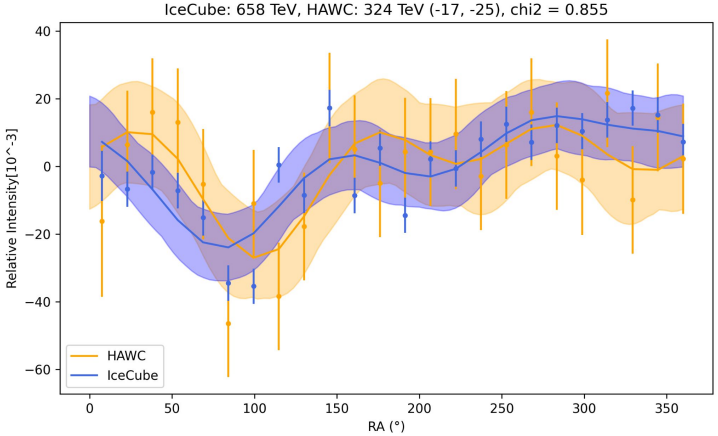
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From Monte Carlo analysis:

HAWC 35.65 TeV -> IceCube 14.5 TeV
HAWC 69.9 TeV -> IceCube 30.54 TeV
HAWC 108.5 TeV -> IceCube 60.635 TeV
HAWC 208 TeV -> IceCube 137.3 TeV
HAWC 377 TeV -> IceCube 349.9 TeV
HAWC 739 TeV -> IceCube 900 TeV

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Conclusions so far

- The **best matching Energy Bins** are **consistent** with the **Rigidity assumption**.
- So, **IceCube** and **HAWC** maps for various **energy ranges match better** in **Rigidity** than they do in Energy.
- This **confirms our theory** that **Cosmic Ray Anisotropy is a magnetic process**.

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I presented this at the **Undergraduate Research Symposium** at **UW-Madison** in April 2025!







Challenges I faced in this project

Implementing a **Weighted Kolmogorov-Smirnov Test function** without prior knowledge of it was difficult.

Initial KS Test function took more than **8 hours to run**, so I had to **optimize** it.

Experimenting with **different approaches** like **vectorization** and **restricting the search window** to get **precise energy ranges** was a task.

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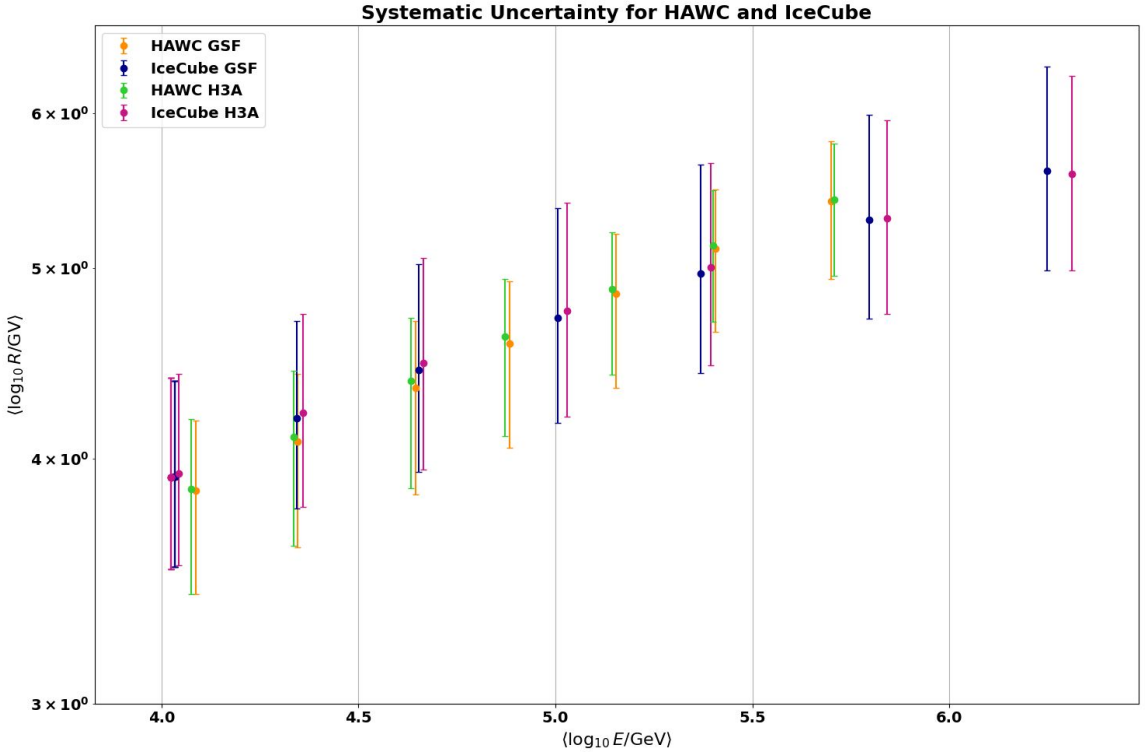
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So, what am I working on right now?



Statistical and Systematic Uncertainty analysis



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- The **preliminary 6 sky-maps** were made with **new HAWC bins** but **old IceCube bins**.
- We still need to **generate new IceCube bins** with the **energy cuts** we got from the **simulation analysis**.

Rerun the **simulation analysis** with an **additional zenith angle cut** to consider **events** only in the **overlapping region.** (For both observatories; already done)

→

Get **new IceCube energy cuts** using a **weighted KS Test** function and submit jobs to get the **real data energy bins**. (On this step right now)

→

Run the code again to **generate new all sky maps** using a **Chi-squared Test**.

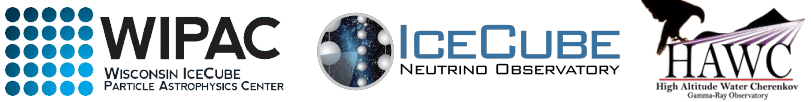
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We are planning to publish two papers for this analysis!

1. HAWC only
2. HAWC + IceCube Cosmic Ray Anisotropy



Thank you!
Questions?





